



NATIONAL UNIVERSITY
OF COMPUTER AND EMERGING SCIENCES
CHINIOT - FAISALABAD CAMPUS



Programming Fundamentals Lab #14

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FAST School of Computing

Instructions:

1. Make a word document with the convention "SECTION_ ROLLNO_L2" and put all your source code and snapshots of its output in it.
2. Plagiarism is strictly prohibited, if you take a code snippet off the internet, mention its reference.
3. Do not discuss solutions with one another.
4. Make a folder (named: 22FXXXX_Section) with all your CPP files and Word file.
5. Submit word file and source file's folder on GOOLE CLASSROOM.

EXAMPLE

C++ program to print a cString.

```
#include <iostream>
#include <cstring>

using namespace std;

int main()
{
    char str[] = "Hello, world!";

    for (int i = 0; i < strlen(str); i++)
    {
        cout << str[i] << endl;
    }

    return 0;
}
```

Task 01:

Write a program that reads a string from the user and outputs it in reverse order using CStrings.

Task 02:

Write a program to reverse the words in a given string *s* while preserving the order of the words. For example, if *s* = "the sky is blue", then your program should output "blue is sky the". You can assume that *s* contains only spaces and alphanumeric characters.

Task 03:

Write a program that reads two strings from the user and outputs whether one is a substring of the other or not using CStrings.

Example

Input

Str1: "television"

Str2: "lev"

Output: Yes Str2("lev") is a substring of Str1("television")

Task 04:

Given a string *s* consisting of words and spaces, return the length of the last word in the string. A word is a maximal substring consisting of non-space characters only.

Input: *s* = "Hello World"

Output: 5

Explanation: The last word is "World" with length 5.

Task 05:

Given an input string *s* and a pattern *p*, implement regular expression matching with support for '.' and '*' where:

'.' Matches any single character.

'*' Matches zero or more of the preceding element.

The matching should cover the entire input string (not partial).

Input: s = "aa", p = "a"

Output: false

Explanation: "a" does not match the entire string "aa".

Input: s = "aa", p = "a*"

Output: true

Explanation: '*' means zero or more of the preceding element, 'a'. Therefore, by repeating 'a' once, it becomes "aa".

Input: s = "ab", p = ".*"

Output: false

Explanation: ".*" means "zero or more (*) of any character (.)".

Task 06:

Given two strings s1 and s2, write a function to determine if s2 is a rotation of s1. For example, "waterbottle" is a rotation of "erbottlewat". Both strings should be taken as an input from the user.

"Optimism is an occupational hazard of programming: feedback is the treatment." **Kent Beck**